

Defect Log and Quality Metrics

Course: Software Testing and Validation

1. Defect Log

Defect ID	Description	Steps to Reproduce	Expected Result	Actual Result	Severity	Priority	Status
DEF-001	30-day duration throws error	1. Book vehicle for exactly 30 days.	Booking succeeds (Total price calculated)	Validation error thrown: "Maximum rental period is 30 days."	High	High	Closed
DEF-002	Missing 'Premium Member' checkbox on form	1. Navigate to booking form.	Checkbox for Premium is visible.	Checkbox is missing from HTML template.	Medium	Medium	Closed
DEF-003	Django ORM Async error in Playwright	1. Run pytest test_system.py	Tests pass	SynchronousOnlyOperation error thrown	Critical	High	Closed
DEF-004	Typos in vehicle list template	1. View vehicle list.	Headers spelled correctly	"Mkae" instead of "Make"	Low	Low	Closed

Defect Lifecycle Evidence

All defects followed the same lifecycle: **New -> Assigned -> In Progress -> Retest -> Closed.**

Defect ID	Lifecycle evidence
DEF-001	New during boundary testing; assigned to the business-logic owner; fixed; retested with TC_Dur_03; closed.
DEF-002	New during UI inspection; assigned to the template owner; fixed; retested in the booking journey; closed.

Defect ID	Lifecycle evidence
DEF-003	New during system-test execution; assigned to the test/integration owner; fixed; Playwright suite rerun; closed.
DEF-004	New during UI inspection; assigned to the template owner; corrected; view test rerun; closed.

2. Quality Metrics

Metric	Calculation / Formula	Result	Interpretation
Defect Density	(Total Defects Found) / (Size of Application - e.g. KLOC or Function Points). Assuming ~0.5 KLOC. (4 / 0.5)	8 defects / KLOC	The codebase is very small, so the density appears artificially high, but 4 total defects is well within acceptable limits for an initial prototype.
Defect Removal Efficiency (DRE)	(Defects found pre-release) / (Total defects found pre-release + escaped defects) * 100. Assuming 0 escaped defects currently. (4 / (4+0)) * 100	100%	All known defects were caught by our automated test pyramid and fixed before code was merged to the main branch.
Code Coverage	Measured with <code>pytest --cov=rental --cov-branch</code> .	94% total; 100% branch coverage for rental/business_logic.py	The core business logic exceeds the 80% branch target. The remaining uncovered lines are mainly the seed command and defensive model/view paths.